Revision 1.0

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Dept. of Electrical, Electronic & Computer Engineering
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Mobile Robots

AUTO4508

Group Project	**Autonomous Navigation**	**weeks 2-12**

GROUPS: Form groups of 4 students

EQUIPMENT: Pioneer 3-AT Outdoor Mobile Robot Platform
Industrial Linux PC with touch screen display
IMU: Phidget Spatial 3/3/3
Camera: Stereo Camera OAK-D V2
Lidar: SICK 2D-Lidar TIM781-2174101 or richbeam lakis1
Software: ROS2

Groups will share the pioneer robots located in the robotics lab.

Part 1 – Simulation (due in week 6) - **Individual**

It is important to recognise the significance of simulation systems in the development of a robot and its software stack. As an individual you will need to adapt the basic simulation folder provided to drive autonomously in an unknown environment while avoiding stationary obstacles. You may use existing packages but you will need to be able to explain how they work and why you chose them.

Tasks to complete

1. Finish the pioneer urdf and ensure it can drive using teleop from ROS2 – a starting guide has been given in the project file.
2. Update the world sdf file to look like James oval. Include several obstacles such as cones and buckets.
3. Add on any relevant sensors that may be needed to drive autonomously. The sensors should not be idealised but follow the limitations of real-world sensors.
4. Set up the ROS2 stack to drive autonomously.
5. Create your own local controller that can drive to a given waypoint ending in the correct position and orientation.
6. You should be able to generate a path of multiple waypoints.
7. Avoid previously unseen static obstacles along the path.
8. Build a map and record the data of the driven path.

Part 2 – Basic navigation (due in week 8) - **Group**

Now it is time to adapt you simulation for the real world.

Tasks to complete

1. Drive the robot along a path, specified through a number of given GPS way-points. The robot has to visit each waypoint before returning to its starting position.
2. Each waypoint is marked by an orange traffic cone. Whenever a waypoint has been reached (within 1-2 meters), the robot must take a photo of the marker object and then head towards the next waypoint. Always leave this marker to the robot's **right** side.
3. Between the 1st and 2nd waypoint there will be a number of cones at unknown intervals, you will need to weave through these cones.
4. At each waypoint, an additional coloured will be in the vicinity, but at an unspecified distance. Identify the shape of the object, record a photo of it, and calculate its distance from the waypoint marker.
5. A summary of the journey should be presented on the completion of driving.
6. Use the Lidar sensor to avoid collisions with markers, objects and any other stationary or moving obstacles, such as walls, vehicles, people, bikes, etc.
7. For safety reasons, implement a Bluetooth link between the robot's on-board PC and a gamepad controller:
    a. 'X' button enables automated mode.
       In automated mode, use the back pedals as a dead-man switch.
       If released, the robot has to stop.

b. Button ‘O’ enable manual mode (disable automated mode).
   In manual mode, the steering controls can be used to manually drive the robot forward/backward and left/right.

Part 3 – Mapping and discovery (due in week 12) - **Group**

Tasks to complete

Your robot has landed on an unknown surface, and it must explore the area and log any details for the arrival of future modules. In order to safely deploy the modules a 15x15 meter area around the initial landing place needs to be searched. Your robot must be able to identify certain colour markers noting the location and colour. There are several areas of interest that will be marked as waypoints and after the initial exploration the robot will need to inspect them a second time as quickly as possible. Your team will need to build a simulation as well to demonstrate the robots capabilities before real world testing can begin.

1. The robot should maintain the capabilities from part 1 while extending the functionality for below.
2. From a set home position you need to explore an unknown area, mapping it as you drive.
3. Within the unknown environment are a number of hand drawn Greek letters. These numbers will be located randomly around the environment on different surfaces at approximately knee height. You will need to use image recognition to determine the number and note its location.
4. You should take photos and note the location of any yellow or red colour obstacles in the area as these are of special interest to the team back home.
5. Avoid collision with moving obstacles and record an incident if an emergency stop is needed (an emergency stop should occur if a moving object comes within 1m of the robot). Note you don’t need to press the estop, this is a software requirement.
6. If a estop event occurs, save the last 5 seconds of recorded data so that it can be reviewed by the team.
7. Implement a user interface with graphics and text on the robot’s display that always displays the robot’s internal state and its intended actions. During the first phase of this part (mapping phase), print the map, all marker photos and locations to the screen. For the second phase (rapid waypoint driving) display the robots planned path graphically to the screen.
8. Once your first run is completed you will be given 3 waypoints noted by the hand drawn numbers to drive to. Your team must plan the fastest path to each of these locations and the path back to home. Your robot must then drive through the unknown environment a second time to each of these waypoints and return home in the fastest time possible.
9. Your system should record the drives so it can be review again offline.
10. You should separate the two phases (mapping and waypoint driving) into two services that can be run from the press of a button or interface without restarting the software stack.

Getting started

You should get started early on the project, particularly planning out the key components the make your robot run. It is always evident when students start late as the debugging process takes time and this has what hurt past students grades the most. You must use ROS2 software stack but are welcome to use other packages (unless stated otherwise in the design brief) as long as you reference where you got the sources. When working with the actual pioneer robots, make use of the software libraries for OAK-D (stereo camera), sick_scan_xd (Lidar) as well as OpenCV (image processing) which stops you from having to reinvent the wheel. As a suggested first node, try and get the PlayStation controller working using the joy library. The Phidget IMU does have a ROS2 library but making your own node tends to achieve better results.

Note that the robots are designed to **only drive outside** on grass or sand.
For **indoor testing**, you need to wrap all tires with tape to allow the rigid wheels to slip – otherwise the robot's motors will burn out (this already happened in the past)!

Resources

Pioneer:

- ARIA Library: https://github.com/cinvesrob/Aria

Phidget IMU:

- User Guide: https://www.phidgets.com/?tier=3&catid=10&pcid=8&prodid=1204
- Code Samples https://www.phidgets.com/?tier=3&catid=10&pcid=8&prodid=1204

OAK-D Camera:

- DepthAI API: https://docs.luxonis.com/projects/sdk/en/latest/
- Code Samples: https://docs.luxonis.com/projects/api/en/latest/tutorials/code_samples/

SICK Lidar:

- Info/ROS-Driver: https://www.sick.com/fr/en/tim781-2174101/p/p594148

Lakibeam:

- ROS2 driver + info: https://github.com/RichbeamTechnology/Lakibeam_ROS2_Driver

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DEMONSTRATION – 60%

In week 6, you will be asked to demonstrate your individual simulation and asked a series of questions related to making your system run. In week 8 you will need to demo part 2 of the project during your allocated workshop time. On the scheduled presentation day (to be confirmed) at the end of the semester, all groups will give a practical demonstration of their projects and answer the project supervisors' questions re. their implementation.

PRESENTATION – 20%

In week 12 your team will need to present your solution to a panel of experts who will then ask you several questions about your implementation and decision-making process. The presentation should highlight the structure of your software design for both parts 2 and 3 with justification for choices. Your ability to answer the questions plus your contributions to the report and user manual provided may impact on your individual functional performance marks.

SUBMISSION – 20%

Online submission with official coversheet incl. declarations of all team members:

1. Project design report for parts 2 and 3 (*pdf*), which includes
    - Report on which team member did what
    - Software design description
    - Diagrams, photos, screenshots, plots, etc.
    - Include page numbers
    - You need to reference any sources that you used.
    - This should be a document you could hand to another design team, and they could develop the same/similar solution based on your design decisions

      Do NOT include:
    - Program code (snippets are ok for explanation purposes)
2. User Manual (*pdf*)
    - Max 5 pages, **no** Title page
    - This document should be a guide for a general (non-technical) user on how to use all the features of your system.
    - As if it was sold to a customer
        - Think about what you see in other user manuals both bad and good
        - A good suggestion would be to show a family member or someone not familiar with robotics and make sure they understand how to use your system.
3. Source code, clearly marking any imported code with referencing the source.

MARKING

60% Functional Performance, Design, Complexity, Innovation

- 15% Simulation
- 15% Part 2
- 30% Part 3

20% Presentation

10% Project Design Report

10% User Manual